CYBERSECURITY PROJECT REPORT

ACTIVE DIRECTORY SECURITY LAB

Kerberoasting Detection & GPO Hardening

Python • Flask • HTML • CSS • CSV • JSON

Educational prototype using fictional sample data only

1. Abstract

This project demonstrates defensive review of fictional Kerberos service-ticket events and basic Group Policy settings. A local Flask dashboard displays findings, severity levels and remediation suggestions.

2. Objectives

Review sample security records; flag activity matching simple training heuristics; audit selected Group Policy settings; present findings in a dashboard; export a CSV report.

3. Tools and Technologies

Python is used for application logic, Flask for the local web app, HTML/CSS for the dashboard, CSV for sample logs and exported findings, and JSON-style settings for the sample policy audit.

4. Project Modules

Kerberos Activity Review flags high ticket counts and legacy RC4 encryption for investigation. The GPO Auditor checks password length, lockout threshold, firewall, administrator MFA, legacy protocols and Guest account status. The dashboard displays findings and recommendations, while the CSV endpoint exports results.

5. Methodology

The application evaluates fictional sample records using simple training rules. High service-ticket counts and legacy RC4 encryption are treated as indicators for review, not proof of malicious activity. Six sample GPO settings are checked against simplified expectations.

6. Demonstration

Run the Flask app locally, explain the summary cards, review the svc_reports sample flagged for high ticket volume and RC4 encryption, review GPO results, and export the findings CSV.

7. Limitations

Sample data is fictional; rules may produce false positives; the GPO audit is not comprehensive; and this prototype is not intended for production security monitoring.

8. Conclusion

The project demonstrates how basic Python rules can organize defensive event review and configuration checks in a local dashboard. All demonstrations are limited to simulated data.

Safety note: Use security tools only in authorized environments. A flagged event is not proof of compromise.